Jaafoura Mouez

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Profile

Third-year ENSIMAG engineering student in Information Systems (ISI), pursuing a dual degree in Artificial Intelligence. Interested in machine learning, generative and multimodal AI, and applied research engineering.

Seeking an end-of-studies internship (PFE) starting February 2027.

Education

ENSIMAG – Grenoble INP Sept 2023 – Present

Engineering degree, Information Systems (ISI) | Third year; dual degree in Artificial Intelligence

Selected coursework: Generative & Multimodal AI, Mathematical Foundations of Machine Learning, Computer Graphics, Large-Scale Data Management.

EISINE – URCA Sept 2021 – May 2023

CUPGE – intensive undergraduate studies in Mathematics and Physics; ranked first in cohort (major de promotion).

Experience

KRACKOS / LIG Jun 2026 – Jul 2026

R&D Systems & ML Engineering Intern Grenoble, France

- Developed a real-time telemetry dashboard for a custom Linux kernel module, combining a C `LD_PRELOAD` library to profile memory allocations with a Node.js/WebSocket backend streaming kernel metrics.
- Integrated ML-based anomaly detection and forecasting for memory usage and page faults; built a Next.js/React interface with Recharts and deployed it behind an Apache reverse proxy with secure WSS.

Notre Grand Bleu Jun 2025 – Aug 2025

Machine Learning Engineer Intern Monastir, Tunisia

- Built a relational database for sea-turtle field data and developed predictive models to estimate turtle populations using Python, Pandas, and PyTorch.
- Cleaned and analyzed ecological data to train and validate the models.

Selected Projects

Multimodal AI Video Creator – Gemini & Veo Python, APIs

- Built a desktop application integrating Gemini for media analysis and Veo for AI video generation across image, audio, and video inputs; added asynchronous generation polling and prompt enrichment for styles and duration.

arXiv Research Companion – RAG Python, Sentence Transformers, ChromaDB, Gradio

- Built a research assistant for arXiv papers using embeddings, vector search, and hybrid BM25 retrieval, with LLM-generated insights and an interface supporting incremental updates.

AI Code Review & Improvement System Python, LLM APIs, Ollama

- Designed a multi-agent code-review workflow combining static-analysis tools with LLMs for code improvement suggestions, test generation, and documentation; added automatic analysis on file changes.

Skills & Technologies

Programming: Python, C, C++, Java, SQL, JavaScript/TypeScript

ML: PyTorch, Pandas, NumPy, Scikit-learn, predictive modeling, anomaly detection

Generative AI: LLM APIs, RAG, embeddings, semantic search, Sentence Transformers, Hugging Face, Gemini, Veo

Tools & Systems: ChromaDB, Gradio, Ollama, Pydantic, Linux, Bash, Git, Node.js, React, Next.js, WebSockets, SQL databases

Languages & Certifications

Languages: English (TOEIC 845), French (TCF C1), Arabic (native), Spanish (conversational)

Certifications: Google Cloud – Introduction to Generative AI; IBM – Introduction to Machine Learning